

General Chemistry 2 – Final Exam (Winter 2022)

Chem120 Sections: 001/002

26th April 2022; 2:00 pm to 5:00 pm

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Assoc. Examiner: Prof. Samuel Sewall

The exam is a closed-book exam. Calculators (non-programmable) are permitted. Crib sheets are not permitted.

AUTOMATED MATCHING GRID PLACEHOLDER

Instructions:

- This booklet has **9 pages (double-sided)** including the cover page.
- Please write legibly and dark enough for a scanner to read. You may use pen or pencil
- **Answer ONLY on the sheets that have a QR code.** Any other sheets will be discarded.
- If you need extra paper to complete a question, use the extra workspace **with the QR-code** at the end of this booklet. Indicate in the appropriate question that the marker should look at Page # for the full answer.
- **Do not write on the QR codes** at the top of the page or your page will not be graded.
- **Do not write beyond the margins and the diagonal line in the top corners.**
- This examination is PRINTED ON BOTH SIDES of the paper
- If you have a question or concern about the exam, please write it on the cover sheet. All questions will be addressed *after* the exam

ALL SHEETS (INCLUDING SCRAP) OF THIS EXAMINATION PAPER MUST BE RETURNED

Chem120 Winter 2022 Final Exam

For all calculations – show your work including equations used. You do not need to show units in the calculations but make sure to show your final answer in the correct units.

Q1 (8 points) Show complete calculations

A) (4 pts) (4 sig. figs.)

A sample of xenon (Xe) gas effuses through a small opening into a vacuum chamber in 142.0 s. A sample of an unknown hetero-diatomic gas diffuses into the vacuum chamber in 55.43 s under identical conditions of pressure, temperature, and concentration as for the Xe. Mass spectrometry measurements identified that hydrogen (H) is one of the atoms in the hetero-diatomic gas. Determine the identity of the unknown hetero-diatomic gas. Justify with calculations.

B) (4 pts) (3 sig. figs.)

A reaction has a rate constant $k = 1.579 \times 10^3 \text{ s}^{-1}$ at a temperature of 350.00°C , and an Arrhenius constant of $1.00 \times 10^{12} \text{ s}^{-1}$. If the enthalpy of the reaction is $\Delta_r H = +20.0 \text{ kJ/mol}$, calculate the activation energy of the reverse reaction in units of kJ/mol.

Q2 (5 pts)

A sample of methane gas (CH_4) having a volume of 2.50 L at 25.00 °C and, a pressure of 1.65 atm was mixed with a sample of oxygen gas (O_2) having a volume of 35.0 L, a temperature of 31.00 °C, and a pressure of 1.25 atm. The mixture was ignited to initiate complete combustion of the methane to form carbon dioxide (CO_2) and water (H_2O).

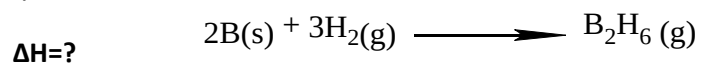
A. (1 pt) Write the balanced combustion chemical reaction

B. (4 pts; 3 sig figs) Calculate the volume of CO_2 formed at a pressure of 2.50 atm and a temperature of 125.00 °C

Q3 (5 pts; 3 sig. figs.)

Boron hydrides are highly reactive compounds and diborane (B_2H_6) has been proposed as a possible rocket fuel by NASA. Calculate the enthalpy change (ΔH) for the synthesis of diborane using the reaction data below. Show your full reactions and thermochemistry calculations for full points value on this question.

Synthetic reaction for diborane:



Thermochemistry reaction data:

- a) $2B(s) + 3/2 O_2(g) \longrightarrow B_2O_3(s)$ $\Delta H = -1273 \text{ kJ}$
b) $B_2H_6(g) + 3O_2(g) \longrightarrow B_2O_3(s) + 3H_2O(g)$ $\Delta H = -2035 \text{ kJ}$
c) $H_2(g) + 1/2 O_2(g) \longrightarrow H_2O(l)$ $\Delta H = -286 \text{ kJ}$
d) $H_2O(l) \longrightarrow H_2O(g)$ $\Delta H = +44.0 \text{ kJ}$

Q4 (10 points)

A) 5 points; 4 sig. figs.

A solid sample of polymer ($m = 2.500 \times 10^{-3} \text{ g}$) is dissolved in enough water to make 1.000 mL of solution. If the osmotic pressure of this solution is $3.000 \times 10^{-3} \text{ bar}$ at 25°C , what is the molar mass of the solid polymer assuming that it is a non-electrolyte?

B) 5 points

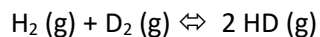
Further research showed that your polymer was actually a strong electrolyte under the pH conditions of the experiment, consisting of a polymer anion (PI^{3-}) counterbalanced by 3 Na^+ ions. In solution the following process would occur:



Is the molar mass you estimated in part A, for the solid polymer, too high or too low? What is the proper correction factor? You can assume the Na_3PI is fully dissociated.

Q5 (10 points)

Hydrogen (H_2) and its isotope Deuterium (D_2) can react to form Hydrogen Deuteride (HD):



A) (4 pts; 3 sig figs) If $K_p = 3.18$ at 273 K; and $K_p = 3.46$ at 383 K; What is ΔH_r° (kJ/mol)?

B) (3 pts) What should the magnitude of the entropy change for this reaction be- large and positive, large and negative, or close to zero. Briefly explain.

C) (3 pts) If a system is prepared with 760 torr of HD, 380 torr of H_2 and 350 torr of D_2 which direction will the reaction go to reach equilibrium at 383 K? Explain your answer.

Q6 (10 points)

A) (5 pts; 3 sig figs)

K_b of a base, $C_5H_{10}NH$, is 5.90×10^{-3} . What is the pH of a 0.500 M solution of $C_5H_{10}NH_2Cl$ at 298 K? Show your work. (To simplify, consider $C_5H_{10}NH$ as B (base) and $C_5H_{10}NH_2Cl$ as BHCl)

B) (5 pts)

At a certain temperature (X), the ionization constant for pure water is 2.75×10^{-14} .

I. Calculate the $[H_3O^+]$ and pOH of pure water at this temperature. Show your work.

II. If the auto-ionization of water is an endothermic process, which of the following (i, ii, or iii) could this temperature (X) be? Justify your answer.

- i. $20^\circ C$ ii. $25^\circ C$ iii. $30^\circ C$

Q7 (10 points)

Using the data below, answer the following questions (Assume $T = 298\text{ K}$)

Compound	pK_b	pK_a
H_2X		4.95
HY		7.12
HZ		10.59
B	9.79	

A) 3 points

Among the given compounds and their conjugates, which is the strongest acid and strongest base? Explain your reasoning

B) 3 points

Which conjugate acid(s)/base(s) combination would be an appropriate buffer system for performing an experiment at $pH = 6.8$. Explain your reasoning.

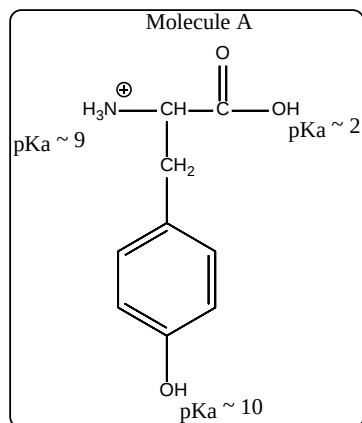
C) 4 points

Two sodium salt solutions of 0.1 M NaY and 0.1 M NaZ are prepared. Which would have a lower pH ? Justify your answer.

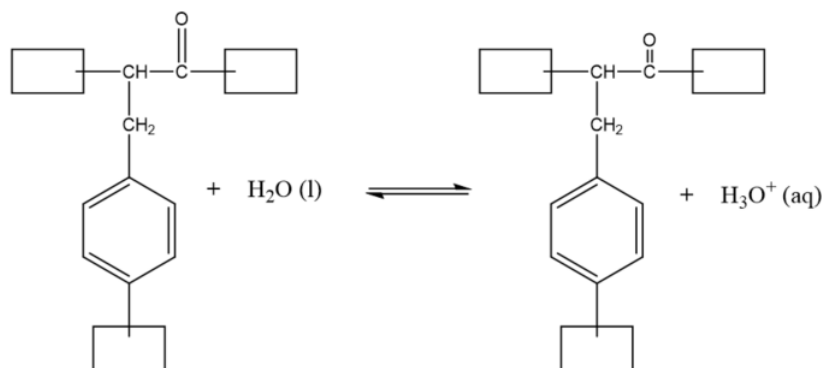
Q8 (11 points)

A) (5pts) The three acidic hydrogens and their pK_a s are marked for the molecule A (left box).

In the boxes provided (in the chemical equation), write down the three acidic groups ($-\text{OH}$; $pK_a = 2$, $-\text{NH}_3^+$; $pK_a = 9$, and $-\text{OH}$; $pK_a = 10$) in their appropriate acidic or basic forms, for the reactant and the product, for the chemical equation representing the *third* acid-dissociation constant, $K_{a,3}$.



Chemical Equation representing $K_{a,3}$



B) (6 pts; 3 sig figs) 100 mL buffer solution is prepared with 0.1500 M Na_2HX and 0.1000 M NaH_2X . What is the change in the pH of the buffer if 60.0 mL of 0.0100 M HCl is added to the buffer?

pK_a of H_2X^- is 6.92. Show calculations, including pH values before and after addition of HCl.

Blank area for calculations.

Q9) 16 points (Assume $T = 298\text{ K}$)

25.00 mL of a 0.1500 M solution of a weak acid ($K_a = 2.200 \times 10^{-5}$) is titrated with 0.1500 M strong base, NaOH.

A) (5 pts; 4 sig figs) Calculate the pH of the solution after addition of 5.000 mL of NaOH. Show your calculations

B) (5 pts; 4 sig figs) Calculate the pH of the solution after addition of 40.00 mL of NaOH. Show your calculations.

C) (6 pts) Which among the following indicator(s), if any, would you choose for this titration (Circle the correct choice(s))? Explain your choice. Justify your choice(s) with calculations.

Thymol B ($pK_{a1} = 1.65$ and 8.9)

or

Bromophenol B ($pK_a = 3.85$)

or

Phenolphthalein ($pK_a = 9.4$)

or

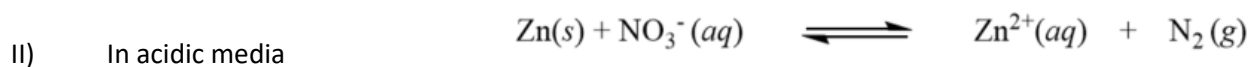
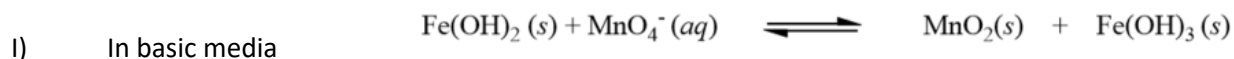
Alizarin Y ($pK_a = 11.2$)

or

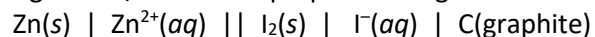
Methyl R ($pK_a = 4.95$)

Q10) 15 points

A. (9 pts) Balance the following equations using smallest whole number coefficients. Show half-reactions and the final fully balanced equation using smallest whole number coefficients.

**B) (6 pts)**

A galvanic/voltaic cell prepared using zinc and iodine has the following cell notation.



Write down the reaction occurring at the anode, the cathode, and correctly balanced spontaneous reaction.

Q11) 10 points

A) 5 points

A solution is prepared by dissolving 45.0 g of NiSO_4 in water. What current would be needed to deposit all of the nickel in 5.0 hours?

B) 5 points

A voltaic cell consists of a Ag/Ag^+ electrode ($E^\circ = 0.80 \text{ V}$) and a $\text{Fe}^{2+}/\text{Fe}^{3+}$ electrode ($E^\circ = 0.77 \text{ V}$) with the following initial molar concentrations: $[\text{Fe}^{2+}] = 0.30 \text{ mol L}^{-1}$; $[\text{Fe}^{3+}] = 0.10 \text{ mol L}^{-1}$; $[\text{Ag}^+] = 0.30 \text{ mol L}^{-1}$. What is the E_{cell} for the voltaic cell? Comment on the spontaneity of the reaction under standard conditions, and the initial conditions provided in the voltaic cell?

Q12) 10 points

Using the table given below, answer the following questions:

Half reactions	E° (volts)
$A^+ (aq) + e^- \rightarrow A(s)$	1.55
$B^+ (aq) + e^- \rightarrow B(s)$	0.77
$C^{2+} (aq) + 2e^- \rightarrow C(s)$	2.05
$X_2 (g) + 2e^- \rightarrow 2X^- (aq)$	-0.88
$D^{2+} (aq) + 2e^- \rightarrow D(s)$	-0.24
$E^{2+} (aq) + e^- \rightarrow E^+(aq)$	1.21

A) (5 pts) You need to construct a replacement for a battery that operates between 1.0 to 1.5 V. Construct one possible galvanic/voltaic cell (give the overall reaction and their standard potential value) that could be used as a replacement for the battery. Explain your choice

B) (2 pts) Which reagent is the strongest oxidizing agent? Justify your choice.

C) (3 pts) Which reagent(s), if any, will oxidize $E^+(aq)$? Justify your choice.